Collision lab

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Test momentum conservation with measured before-and-after velocities.

SUCCESS CRITERIA - I can...

1. Record masses, signs and velocities consistently.
2. Compare total momentum before and after.
3. Discuss external impulse and measurement variation.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “trial” and “momentum” to explain your answer. Sentence starters are optional.

trial: one run or repetition of a test

momentum: mass times velocity, including direction

external: acting from outside the chosen system

01 Warm-up: bring back an earlier idea

The same object stops from the same speed, but the stopping time is **3** times the original. **Compare average stopping-force magnitudes**.

Word help: average = one value representing an interval or a set of values

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $p_{total} = m_1v_1 + m_2v_2$; $\Delta p_{total} = p_{total,\,f} - p_{total,\,i}$. **Compare just before**/after collision. Use approved carts; **record actual trials separately** from practice data.

Worked reasoning: Supplied trials give total momenta **6.0** before and **5.8 kg·m/s** after. The change is **-0.2 kg·m/s**. **Compare this** with measurement uncertainty before attributing it to a new physical effect.

Your turn: Draw a before-and-after momentum comparison and **name one possible external interaction**.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Before | **After**

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Supplied case: cart A is **0.50 kg** at **+0.40 m/s**; B is **0.50 kg** at rest. After sticking, both move at **+0.20 m/s**. **Find total momentum before** and after.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found ... [unit]. This value means ... in this situation.

Cart	Before (m/s)	After (m/s)
A: 0.50 kg	+0.40	+0.20
B: 0.50 kg	0	+0.20

Marzano check - circle: ① ② ③

04 Apply the physics

Two **1 kg** lab carts stick together. Before contact their velocities are **+2** and **0 m/s**; afterward both move at **+1 m/s**. **Complete the table**. **Compare total momentum before** and after and **write the statement the data** support.

Sentence starter: Both ... . They differ in ... because ... .

Cart	Mass (kg)	v before (m/s)	v after (m/s)
A	1	2	1
B	1	0	1

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

1. Design a table for three trials with masses and signed velocities before/after.
2. **Name one possible external impulse** and one measurement limitation.

Sentence starter: My design includes … . It does what is needed by … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A launch has v_x=**6 m/s** and v_y, i=**40 m/s** upward. Use g=**10 m/s²**. **Find time** to the top and total time back to launch height.
Label separate horizontal and vertical motion.

Sentence starter: I found … [unit]. This value means … in this situation.



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

07 Exit check

Why must velocities be signed when carts approach each other?

Sentence starter: I think … . I can point to …, which shows …, so … .

Object / trial	Mass	Velocity	Momentum
A	____	____	____
B	____	____	____

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

Group A has momentum change **0.1±0.2**; group B has **0.8±0.2 kg·m/s**. Which result is consistent with zero under the stated bounds?

Sentence starter: I notice … . This supports … because … .

Object / trial	Mass	Velocity	Momentum
A	____	____	____
B	____	____	____

Marzano check - circle: ① ② ③